Chapter 9

Finite Volume Discretization

9.1 One-dimensional setting: The Basic Ideas

Consider the Cauchy problem

$$\begin{aligned} u_t + (f(u))_x &= 0, \qquad (x,t) \in \mathbb{R} \times (0,\infty) && (9.1)\\ u(x,0) &= u_0(x). && (9.2) \end{aligned}$$

Partition the real line into non-overlapping intervals, such that $\mathbb{R} = \bigcup_{i\in\mathbb{Z}} I_i$, and we define $I_i = [x_{i-\frac{1}{2}}, x_{i+\frac{1}{2}}]$. Denote the centroid of an interval $x_i := \frac{1}{2}(x_{i-\frac{1}{2}} + x_{i+\frac{1}{2}})$, and integrate (9.1) over I_i to obtain

$$\int_{I_i} u_t + (f(u))_x \, dx = 0. \tag{9.3}$$

The volume of an interval is defined

$$h := \int_{x_{i-\frac{1}{2}}}^{x_{i+\frac{1}{2}}} dx = x_{i+\frac{1}{2}} - x_{i-\frac{1}{2}}.$$

(Note that with this definition h could be different in each interval. For simplicity we assume that h is constant.) Next we introduce the *volume average*

$$u_i(t) := \frac{1}{h}\int_{I_i} u(x,t)\, dx,$$

and write (9.3) as

$$\begin{aligned} h\frac{du_i}{dt} + \int_{I_i} (f(u))_x \, dx &= 0\\ \Leftrightarrow \quad h\frac{du_i}{dt} + f(u(x_{i+\frac{1}{2}})) - f(u(x_{i-\frac{1}{2}})) &= 0. \end{aligned}$$

We define a numerical scheme by introducing as the unknowns the volume averages u_i. To get a pointwise definition of the numerical solution, we could then use a step function

$$u_h := \sum_{i\in\mathbb{Z}} u_i \chi_i, \tag{9.4}$$

where $\chi(I_i)$ is the *characteristic function* defined on the interval I_i:

$$\chi_i(x) = \begin{cases} 1\,, & x \in I_i \\ 0\,, & \text{otherwise}\,. \end{cases} \tag{9.5}$$

A Taylor series argument reveals that an approximation of the type (9.4) cannot be better than first-order accurate in the point-wise sense. Alternatively, we could attempt a higher order polynomial reconstruction from the volume averages. Of course, whether this makes sense, depends on how well we approximate the volume averages.[1] For now, it will suffice to work with the volume averages directly.

We need some kind of method at the interval interfaces to approximate the flux $f(u(x_{i+\frac{1}{2}}))$. For example, we may express the flux function there in terms of the volume averages in adjacent intervals. To this end, define a function $\hat{f}$, such that

$$\hat{f}_{i+\frac{1}{2}} = \hat{f}(u_i, u_{i+1}) \approx f(u(x_{i+\frac{1}{2}})),$$

We shall give specific examples for such an approximation below. For now we define:

Scheme 9 (1D Finite Volume Scheme) For $i \in \mathbb{Z}$ define

$$h\frac{du_i}{dt} + \hat{f}_{i+\frac{1}{2}} - \hat{f}_{i-\frac{1}{2}} = 0, \tag{9.6}$$

$$u_i(0) = \frac{1}{h}\int_{x_{i-\frac{1}{2}}}^{x_{i+\frac{1}{2}}} u_0(x)\,dx. \tag{9.7}$$

We can define a fully discrete scheme using any discretization of the time derivative. For example, we may use a simple forward difference to produce

$$\frac{u_i^{n+1} - u_i^n}{\tau} + \frac{1}{h}(\hat{f}^n_{i+\frac{1}{2}} - \hat{f}^n_{i-\frac{1}{2}}) = 0, \qquad i \in \mathbb{Z}\,,\ n = 0, 1, \ldots, \tag{9.8}$$

where the superscript n indicates, as usual, evaluation at discrete time instance $t^n = n\tau$.

For the schemes we discuss here, it makes no sense to evaluate the integral (9.7) exactly. Instead of using exact integration, the midpoint rule $u_i(0) \approx u_0(x_i)$ may be used.[2]

Example 9.1 Consider the linear advection equation

$$u_t + au_x = 0.$$

Setting $f(u) := au$, we can write the advection equation as a conservation law

$$u_t + (f(u))_x = 0.$$

Recall the upwind scheme (6.10). We may write that scheme in the form (9.8) if we set

$$\hat{f}_{i+\frac{1}{2}} = \frac{a}{2}(u^n_{i+1} + u^n_i) - \frac{|a|}{2}(u^n_{i+1} - u^n_i). \tag{9.9}$$

It should be noted that whether we view the upwind scheme for this problem as a finite difference scheme or as a finite volume scheme is a matter of interpretation. Also, recall that the upwind scheme is only first order consistent, so clearly one can evaluate the initial conditions using the midpoint rule.

[1] One has to carefully distinguish between the accuracy of u_h, as defined by (9.4), and the accuracy of the volume averages u_i. In this chapter, we restrict ourselves to first order accurate schemes, i.e. the scheme for the volume averages u_i is first order consistent (see below). If the volume averages are known to higher accuracy, then a pointwise approximation of higher order can be generated using interpolation.

[2] The midpoint rule is second order accurate. Higher order approximation of the initial volume averages makes sense only if the overall accuracy of the scheme, i.e. the order of consistency of the volume averages u_i is higher than that. For the schemes we discuss here, this is not the case.

Even if we treat a more general equation, the interpretation of the 1D finite volume method as a finite difference method is straightforward, as the next example shows.

Example 9.2 Consider the general conservation law

$$u_t + (f(u))_x = 0.$$

Approximate this using the finite volume nomenclature (9.8) with a flux function inspired by the linear advection case (9.9),

$$\hat{f}_{i+\frac{1}{2}} = \frac{1}{2}(f(u_{i+1}^n) + f(u_i^n)) - \frac{\alpha}{2}(u_{i+1}^n - u_i^n). \tag{9.10}$$

where α is now some generic "diffusion" coefficient, which for now we assume to be a constant. We can write

$$\frac{u_i^{n+1} - u_i^n}{\tau} + \frac{1}{2h}(f(u_{i+1}^n) - f(u_{i-1}^n)) - \frac{\alpha}{2}\frac{u_{i-1}^n - 2u_i^n + u_{i+1}^n}{h} = 0, \qquad i \in \mathbb{Z}\,,\ n = 0, 1, \ldots,$$

where in the finite difference parlance we interpret the second term as a central difference approximation of $(f(u))_x$, and the right hand side is consistent with

$$\frac{\alpha}{2}\frac{u_{i-1} - 2u_i + u_{i+1}}{h} = \frac{h\alpha}{2}u_{xx} + \mathcal{O}(h^3).$$

We have used diffusive terms before to stabilize central difference approximations for hyperbolic equations, and we may take the general form of the flux function (9.10) as a blueprint for the design of finite volume schemes.

It is easy to see that the scheme with flux (9.10) is overall first order consistent for smooth solutions: If we substitute the exact solution into scheme, we obtain

$$\underbrace{u_t + (f(u)_x)}_{=0} + \mathcal{O}(\tau + h^2) + \frac{h\alpha}{2}u_{xx} + \mathcal{O}(h^3).$$

So far the difference between finite volume methods and finite difference methods has appeared to be largely a matter of interpretation, and we may ask why we should even bother with finite volume schemes. The strength of the finite volume approach is seen in the multidimensional case. The extension of the 1D method to the multidimensional case operating on rather general meshes is straightforward in the finite-volume framework. The finite difference formulation is not so easily extended to the multidimensional case on general meshes[3].

9.2 Multidimensional Extension

The advantage of the finite volume method is the easy extension to multiple dimensions. Consider the d-dimensional scalar Cauchy problem

$$u_t + \sum_{i=1}^{d} (f_i(u))_{x_i} = 0, \qquad (x,t) \in \mathbb{R}^d \times (0,\infty) \tag{9.11}$$

$$u(x,0) = u_0(x)\ . \tag{9.12}$$

[3]The phrase "general meshes" is important here. Extending the finite difference method to simple two-dimensional meshes in space that look like fig. 3.1 is not hard. However, for many problems of practical interest with complex geometry meshes composed of, say, triangles are used. Here the finite volume framework is more convenient.

For a proper discretization in multiple dimensions, we need a more general concept of a grid. To avoid dealing with boundary conditions, we formally consider a partition of $\mathbb{R}^d$ into non-overlapping polyhedral elements T_i. For purposes of illustration we often use $d = 2$, and assume the elements to be triangles.

Consider the volume average

$$u_i(t) := \frac{1}{|T_i|} \int_{T_i} u(x,t)\, dx,$$

where

$$|T_i| := \int_{T_i} dx$$

is the volume of the element, and integrate (9.11) over T_i. We obtain, using the divergence theorem,

$$|T_i| \frac{du_i}{dt} + \int_{\partial T_i} f(u) \cdot n\, ds = 0 \ . \tag{9.13}$$

Here we have defined $f = (f_1, \ldots, f_d)$. For polyhedral mesh elements, the boundary of T_i consists of, say b_i, planar faces. (For general polyhedra, the number of boundary faces may vary from cell to cell. For triangles the boundary is given by three straight line segments, called *edges*.) Let the faces comprising ∂T_i be denoted e_k for $k = 1, \ldots b_i$. We have

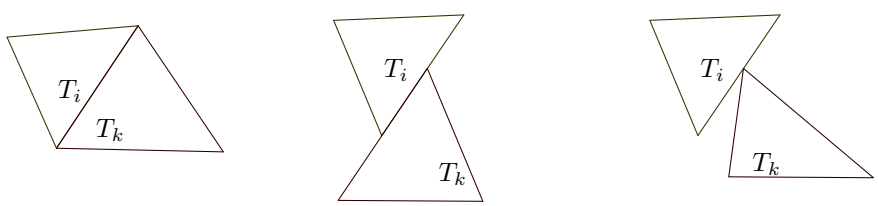
$$|T_i| \frac{du_i}{dt} + \sum_{k=1}^{b_i} \int_{e_k} f(u) \cdot n_k ds = 0,$$

where n_k is the outward pointing normal on face e_k.

We re-write the boundary integral as a sum over the *interfaces* between T_i and neighboring elements,

$$|T_i| \frac{du_i}{dt} + \sum_{k \in \mathcal{N}_i} \int_{S_{ik}} f(u) \cdot n_{ik} ds = 0,$$

where $\mathcal{N}_i := \{k : T_k \text{ is a neighbor of } \mathrm{T_i}\}$. (See Fig. 9.1 for an illustration for the case of triangles.[4]) The interface shared by elements T_i and T_k is denoted by S_{ik}, while n_{ik} is the outward pointing normal (i.e. pointing from T_i to T_k).



(a) Triangles T_i and T_k are neighbors

(b) Triangles T_i and T_k are neighbors

(c) Triangles T_i and T_k are *not* neighbors

Figure 9.1: Illustration of different configurations of triangles with non-empty intersection, and non-overlapping interior. A neighbor of T_i is a triangle that shares an edge, or a segment of an edge, with T_i. If a triangle just shares an isolated point (a vertex) with T_i it is *not* a neighbor.

As in the one-dimensional case, we define a numerical scheme by considering the volume averages as the unknowns. The values on the edges ∂T_i are thus double-valued, and this necessitates again a suitable approximation of the flux. We may approximate the integral over the

[4]It is worth pointing out that while for the purposes of this section there is no need to specify in more detail exactly what kind meshes we consider, in later sections, we shall see that some numerical schemes do *not* allow configurations as in 9.1b. (Instead we will stipulate that if two triangles are neighbors, the shared interface must be an edge of both triangles, as in 9.1a.) Furthermore, when we carry out convergence analysis, we need to specify more clearly in what way one may "let the mesh size go to zero".

face with a midpoint quadrature rule and express $f(u) \cdot n$ as a function of the volume average of neighboring elements. We obtain the scheme

$$|T_i| \frac{du_i}{dt} + \sum_{k \in \mathcal{N}_i} \hat{f}(u_i, u_k; n_{ik}) |S_{ik}| = 0 \ , \qquad i = 1, 2 \ldots, \tag{9.14}$$

Here, $|S_{ik}|$ denotes the area of the interface S_{ik}. This is the classical finite-volume scheme in semi-discrete form. The numerical flux may be approximated by a variety of different methods. For example, we may use

$$\hat{f}(u_i, u_k; n_{ik}) := \frac{1}{2}(f(u_k) + f(u_i)) \cdot n_{ik} - \frac{\alpha_{ik}}{2}(u_k - u_i).$$

which is inspired by the flux we used as a blueprint in the one-dimensional case (cf. eq. (9.10)).

Equation (9.14) is a system of nonlinear ordinary differential equations (ODE). The simplest fully discrete finite-volume scheme is obtained by using a one-step forward discretization in time, i.e.

$$\frac{d}{dt} u_i \approx \frac{u_i^{n+1} - u_i^n}{\tau},$$

where u_i^n denotes the numerical approximation of the cell avarage at time t^n, which leads to

$$u_i^{n+1} = u_i^n - \frac{\tau}{|T_i|} \sum_{k \in \mathcal{N}_i} \hat{f}(u_i^n, u_k^n; n_{ik}) |S_{ik}|. \tag{9.15}$$

9.3 Stability Analysis

We do not assume that the numerical flux functions are smooth. But we will assume that they are at least Lipschitz continuous. In addition, we note a few basic definitions that will be useful in the analysis below.

Definition 9.1 *A numerical flux function $\hat{f}$ is called* conservative *if it satisfies*

$$\hat{f}(u, v; n) = -\hat{f}(v, u; -n) \ . \tag{9.16}$$

Definition 9.1 ensures that flux contributions from interior edges cancel in the summation of Eq. (9.14) over an arbitrary number of connected mesh elements. This is a discrete analog of the property of the integral conservation laws that the rate of change for arbitrary control volumes be given by the flux across the boundary[5].

The numerical flux was introduced as an approximation to $f(u) \cdot n$ at interfaces between two mesh elements. This is necessary, because the numerical solution is discontinuous across these interfaces, and thus double-valued. The numerical flux $\hat{f}$ is a function of both these values. It makes sense that the numerical flux should give back the exact flux function if both arguments are the same. This leads to the following notion of consistency:

Definition 9.2 *We call a numerical flux function $\hat{f}$* consistent *if it satisfies*

$$\hat{f}(u, u; n) = f(u) \cdot n \tag{9.17}$$

[5]Recall the derivation of the compressible Euler equations in Chapter 8.1.

Note that, in particular this holds if u is the exact solution. It is easily seen that scheme (9.14) is consistent, if the numerical flux is consistent.

In the following we always assume that the numerical flux functions which we use are conservative and consistent. These ingredients are essential. A third property is very helpful for the analysis of finite-volume schemes, although it is not always verified in practice.

Definition 9.3 *A numerical flux function $\hat{f}(u, v; n)$ is called* monotone *if it is monotonically increasing / decreasing in the first and second input argument, respectively. In the case of differentiability we have*

$$\frac{\partial \hat{f}}{\partial u} \geq 0 \ , \qquad \frac{\partial \hat{f}}{\partial v} \leq 0 \ . \tag{9.18}$$

Before going to the fully discrete scheme, we note the consequence of the monotonicity definition. Begin by writing scheme (9.14) as

$$\begin{aligned} \frac{du_i}{dt} &= -\frac{1}{|T_i|} \sum_{k \in \mathcal{N}_i} \hat{f}(u_i, u_k; n_{ik}) |S_{ik}| \\ &= -\frac{1}{|T_i|} \sum_{k \in \mathcal{N}_i} \left(\hat{f}(u_i, u_k; n_{ik}) - f(u_i) \cdot n_{ik} \right) |S_{ik}| \end{aligned} \tag{9.19}$$

where we have used

$$\sum_{k \in \mathcal{N}_i} f(u_i) \cdot n_{ik} |S_{ik}| = f(u_i) \cdot \left(\sum_{k \in \mathcal{N}_i} n_{ik} |S_{ik}| \right) = f(u_i) \cdot \left(\int_{\partial T_i} n ds \right) = 0,$$

as we integrate over a closed surface. (Also recall that the boundary is composed of planar faces. This explains the last equality.) Note that for a consistent flux, if $u_i = u_k$, the corresponding term in the sum on the right-hand side of (9.19) is zero. Assume now that $u_i \neq u_k$. Then

$$\begin{aligned} \frac{du_i}{dt} &= -\frac{1}{|T_i|} \sum_{k \in \mathcal{N}_i} \frac{\hat{f}(u_i, u_k; n_{ik}) - f(u_i) \cdot n_{ik}}{u_k - u_i} (u_k - u_i) |S_{ik}| \\ &= -\frac{1}{|T_i|} \sum_{k \in \mathcal{N}_i} \frac{\hat{f}(u_i, u_k; n_{ik}) - \hat{f}(u_i, u_i; n_{ik})}{u_k - u_i} (u_k - u_i) |S_{ik}| \\ &= \frac{1}{|T_i|} \sum_{k \in \mathcal{N}_i} C_{ik}(u_h)(u_k - u_i) \ , \end{aligned} \tag{9.20}$$

where we have used the consistency property, Definition 9.2. Finally, the monotonicity property, Definition 9.3 ensures that

$$C_{ik} = -\frac{1}{|T_i|} \underbrace{\frac{\hat{f}(u_i, u_k; n_{ik}) - \hat{f}(u_i, u_i; n_{ik})}{u_k - u_i}}_{\leq 0} |S_{ik}| \geq 0 \tag{9.21}$$

If the numerical flux is Lipschitz continuous, the coefficients are bounded from above as well. Furthermore, since right-hand side (9.20) satisfies a Lipschitz condition, the ODE has a unique solution. It is easy to see that the solution verifies a maximum principle, i.e. has the property that local maxima/minima cannot increase/decrease.

The fully discrete case (9.15) can be written

$$u_i^{n+1} = \sum_{k \in \mathcal{N}_i^0} \widetilde{C}_{ik}^n u_k^n, \tag{9.22}$$

where

$$\mathcal{N}_i^0 = \mathcal{N}_i \cup \{i\}.$$

The coefficients $\widetilde{C}_{ik}^n = \widetilde{C}_{ik}(u_h^n)$ are given by

$$\widetilde{C}_{ik}^n = \begin{cases} \frac{\tau}{|T_i|} C_{ik}^n & k \neq i \\ 1 - \frac{\tau}{|T_i|} \sum_{k \in \mathcal{N}_i} C_{ik}^n & k = i \end{cases} \tag{9.23}$$

with C_{ik} defined as in (9.20). Note that

$$\sum_{k \in \mathcal{N}_i^0} \widetilde{C}_{ik}^n = 1, \tag{9.24}$$

which asserts consistency. (We have already noted this in Section (4.3) for the linear case.)

We have seen in section (4.3) that positivity of the coefficients plays a major role in establishing stability in maximum norm. It turns out that this holds for the nonlinear case as well.

Theorem 9.1 *Let scheme* (9.22) *be consistent. Then one has*

$$\max_i |u_i^{n+1}| \leq \max_i |u_i^n| \tag{9.25}$$

if and only if $\widetilde{C}_{ik}^n \geq 0$, *for all* i, k.

Proof We show that $\sum_k |\widetilde{C}_{ik}^n| \leq 1$, for all i is necessary and sufficient for (9.25) to hold. For a consistent scheme, i.e. if (9.24) holds, this condition is equivalent to $\widetilde{C}_{ik}^n \geq 0$.

First, we show $\sum |\widetilde{C}_{ik}| \leq 1$ is sufficient. Indeed, for any i we have

$$\begin{aligned} |u_i^{n+1}| &= \left| \sum_{k \in \mathcal{N}_i^0} \widetilde{C}_{ik}^n u_k^n \right| \\ &\leq \sum_{k \in \mathcal{N}_i^0} |\widetilde{C}_{ik}^n| \, |u_k^n| \\ &\leq \max_i |u_i^n| \sum_{k \in \mathcal{N}_i^0} |\widetilde{C}_{ik}^n| \end{aligned}$$

If $\sum_k |\widetilde{C}_{ik}^n| \leq 1$ holds for all i, then (9.25) follows.

Next, we show $\sum |\widetilde{C}_{ik}^n| \leq 1$ is necessary. We merely have to find one counterexample. Consider the linear case $f(u) = au$. Then $\widetilde{C}_{ik}^n \neq C_{ik}^n(u_h)$, and for fixed i, let $u_k^n = sign(\widetilde{C}_{ik}^n)$ for $k \in \mathcal{N}_i^0$, and $u_k^n = 0$ for $k \notin \mathcal{N}_i^0$. Note that this implies $\max_i |u_i^n| = 1$. We have

$$|u_i^{n+1}| = \sum_{k \in \mathcal{N}_i^0} |\widetilde{C}_{ik}^n|$$

if $\sum |\widetilde{C}_{ik}^n| > 1$, clearly $\max_i |u_i^{n+1}| > 1$. ∎

Theorem 9.1 is a rather general result, and applies to any scheme that can be written in the canonical form (9.22). In particular, for our finite-volume scheme, where the coefficients $\widetilde{C}_{ik}^n$ are given by (9.23), we can derive a time-step restriction that ensures positivity. One has the following theorem

Theorem 9.2 *Assume we are given a discrete finite volume scheme of the form (9.15) with consistent and monotone fluxes. Assume that the time step restriction*

$$1 - \frac{\tau}{|T_i|} \sum_{k \in \mathcal{N}_i} C_{ik}(u_h^n) \geq 0, \qquad \forall i, \tag{9.26}$$

holds for each time increment $[t^n, t^{n+1}]$, where the coefficients C_{ik} are defined in (9.20). Then the scheme is stable in maximum norm, and the following estimate holds:

$$\inf_{x \in \mathbb{R}^d} u(x,0) \leq u_i^n \leq \sup_{x \in \mathbb{R}^d} u(x,0) \ , \qquad \forall i, n \ . \tag{9.27}$$

Proof Using the constraint (9.26) in the fully discrete form, written as in (9.22), we have non-negative coefficients (cf. eq. (9.23). Recall that monotonicity of the numerical flux ensures that the coefficients C_{ik} are non-negative.) Stability then follows from theorem 9.1. Upon applying $\max_i |u_i^{n+1}| \leq \max_i |u_i^n|$ recursively up to the initial conditions, one immediately finds (9.27). ■

Very few ingredients are thus needed in order to establish nonlinear stability of the finite volume scheme on general triangulations. The most succinct summary of the machinery which we have just established is that for consistent and monotone fluxes, we can always ensure stability of the discrete scheme (9.15) by choosing the time step as in (9.26). There are, however, many open points: Firstly, due to the nonlinearity of the equations, establishing convergence to the entropy weak solution and pertaining convergence rates is much more difficult. It has to be stated that no general equivalence theorem between consistency and stability on the one hand, and convergence on the other hand applies in the nonlinear case. Secondly, one may attempt to design higher order consistent schemes. (The present formulation is restricted to first order consistency.) This necessitates a careful interpolation of the volume averages to produce, e. g. a piecewise linear representation of the solution. Also, while monotone fluxes are sufficient to establish stability, it is not a strictly necessary condition. Other conditions can be given as well. For convenience we will stick to monotonicity, however.

9.4 Constructing Schemes

In this section we investigate how, in practice we can design a stable finite volume scheme. From the analysis in the preceding section it is clear that this can be done by constructing monotone fluxes.

Consider the scheme

$$\frac{du_i}{dt} = -\frac{1}{|T_i|} \sum_{k \in \mathcal{N}_i} \hat{f}(u_i, u_k; n_{ik}) |S_{ik}| \tag{9.28}$$

with the flux function

$$\hat{f}(u_i, u_k; n_{ik}) := \frac{1}{2}(f(u_k) + f(u_i)) \cdot n_{ik} - \frac{\alpha_{ik}}{2}(u_k - u_i).$$

We claim the scheme is stable for

$$\alpha_{ik} \geq |\overline{a}_{ik}|, \tag{9.29}$$

where

$$\overline{a}_{ik} = \begin{cases} \frac{(f(u_k) - f(u_i)) \cdot n_{ik}}{u_k - u_i} & u_k \neq u_i \\ f'(u_i) \cdot n_{ik} & u_k = u_i \end{cases} \tag{9.30}$$

With this definition, the flux is consistent and conservative. We may write

$$\begin{aligned}\hat{f}(u_i, u_k; n_{ik}) &= f(u_i)\cdot n_{ik} + \frac{1}{2}(f(u_k) - f(u_i))\cdot n_{ik} - \frac{\alpha_{ik}}{2}(u_k - u_i) \\ &= f(u_i)\cdot n_{ik} + \frac{1}{2}\left(\overline{a}_{ik} - \alpha_{ik}\right)(u_k - u_i) \end{aligned} \tag{9.31}$$

Note that $(f(u_k) - f(u_i))\cdot n_{ik} = \overline{a}_{ik}(u_k - u_i)$ holds by definition (9.30). (If $u_i = u_k$ this holds trivially as long as the coefficients $\overline{a}_{ik}$ remain finite. This is also ensured by (9.30).) Alternatively, isolating $f(u_k)$, we may write the flux as

$$\begin{aligned}\hat{f}(u_i, u_k; n_{ik}) &= f(u_k)\cdot n_{ik} - \frac{1}{2}(f(u_k) - f(u_i))\cdot n_{ik} - \frac{\alpha_{ik}}{2}(u_k - u_i) \\ &= f(u_k)\cdot n_{ik} - \frac{1}{2}\left(\overline{a}_{ik} + \alpha_{ik}\right)(u_k - u_i) \end{aligned} \tag{9.32}$$

If (9.29) holds, it is obvious from eq. (9.31) that $\hat{f}$ is monotonically decreasing in its second argument, and from (9.32) it is clear that it is monotoncially increasing in its first argument. Hence, the flux is monotone in the sense of Definition 9.3. We can write the scheme in the from (9.20), with positive coefficients. To make this explicit, substitute (9.31) into (9.28) to obtain

$$\frac{du_i}{dt} = \frac{1}{|T_i|}\sum_{k\in\mathcal{N}_i}\frac{1}{2}\left(\alpha_{ik} - \overline{a}_{ik}\right)(u_k - u_i)|S_{ik}|.$$

(Note that the first term in (9.31) vanishes in the summation over a closed surface.) Clearly the coefficients $C_{ik} = 1/2\left(\alpha_{ik} - \overline{a}_{ik}\right)|S_{ik}|$ are positive under the constraint (9.29). We can conclude that, for u_i a local minimum (respectively maximum) $\frac{du_i}{dt} \geq 0$ (respectively $\frac{du_i}{dt} \leq 0$). The fully discrete scheme can be made stable under the simple time step restriction as in Theorem 9.2.